TECHNICAL REPORT · PHASE 2 « AUTOMATISER »

CRM and website for Jérémy Galéa (Méthode JG®)

What I added from the CI pipeline and the notes scanner up to milestone M4

Rafik · 15 October 2026 · Branch `feat/phase-2`

Context

The project is a CRM and a public website for Jérémy Galéa, a life coach and SwissWAI client. Both live in one Next.js 16 application (TypeScript, PostgreSQL with Drizzle ORM, Better Auth for login). Phase 1 (the CRM and the website) is merged into `main`.

This report covers what I added since then: the CI pipeline, then Phase 2 up to milestone M4. I also finished M5, so I included it at the end.

Part	What it adds	Status
CI pipeline	Automatic checks on GitHub for every push and pull request	On `main`
M1 · E-mail	Sending, logging, test mode and scheduled jobs	Done
Notes scanner	Photos of handwritten notes become a clean PDF in the client file	Done
M2 · Reminders	E-mails 24 h and 1 h before sessions, private calendar feed	Done
M3 · Booking	Days off, move / cancel page, client booking links, notifications	Done
M4 · Sequences	Editable e-mails and automatic follow-ups	Done
M5 · Questionnaire	Intake questionnaire before the first session	Done

1 CI pipeline (GitHub Actions)

Every push and pull request now runs the same checks on GitHub:

- type check, ESLint with zero warnings allowed, and the unit tests;
- a check that every database schema change comes with its migration;
- migrations and seed data run on a real PostgreSQL 16 service;
- a production build.

I added a second workflow, « Lockfile (Linux) », that regenerates `package-lock.json` on Linux. I needed it because npm on Windows leaves out some optional packages (`@emnapi/*`), which made `npm ci` fail in CI. It runs by itself when `package.json` changes; I download the file and commit it.

2 M1 · E-mail foundation

- E-mails are sent through SMTP (nodemailer), configured only in `.env`. Without credentials the app stays in « test mode »: e-mails are logged, not sent. While testing, a redirect sends every e-mail to my own inbox.
- Every e-mail is recorded in an `email_log` table (queued, sent or failed) and shown in the contact history and in Settings.

- One e-mail layout in Jérémy's brand colours, with a plain-text version, and French texts.
- Scheduled jobs go through one endpoint, `POST /api/cron/run`, protected by a secret. The server's cron will call it every 5 minutes. Each job is written so that running it twice does nothing twice.

3 Notes scanner

Jérémy takes handwritten notes during sessions. From the client file he can now photograph the pages and get a clean PDF:

- OpenCV.js (WebAssembly, so nothing native to install on the server) finds the sheet, corrects the perspective and evens out the background to white, so the result looks like a real scan;
- pdf-lib builds a branded PDF: a cover (title, client, date, session, keywords), then one page per sheet;
- the PDF is saved in the client's documents with its title, keywords and linked session.

> One problem worth mentioning: OpenCV's module behaves like a promise that resolves to itself, so a dynamic `import()` never finished. I load it with `createRequire` instead. A unit test runs the whole pipeline on a generated « photo ».

4 M2 · Reminders and calendar feed

- **Reminders.** An e-mail goes out 24 h and 1 h before every planned session. The job marks sessions with a single `UPDATE … WHERE reminder_24h_sent_at IS NULL RETURNING`, so a second run, or two runs at the same time, never sends a reminder twice. Moving a session resets its reminders.
- **Private links.** I added one table, `access_tokens`, for every link that works without a login (managing a booking, the calendar feed, client booking links, the questionnaire, unsubscribing). Only a SHA-256 hash is stored, and the token itself is an HMAC of the row id with the server secret. A copy of the database alone opens nothing, and links can expire or be revoked.
- **Calendar feed.** A private `.ics` address that Jérémy can subscribe to in Google Calendar, on his iPhone or in Outlook. It only shows the type of session, the first name and initial, and the place, never notes. He can copy, regenerate or disable it in Settings.

5 M3 · Online booking

- **Days off and extra hours.** Jérémy's availability changes from week to week, so there is no fixed holiday calendar. He blocks a day, several days or a few hours (or opens extra hours) whenever he needs to. The booking page updates right away, and he gets a warning if sessions are already planned in that period.
- **Move or cancel page.** It is linked from the confirmation and reminder e-mails and works until a deadline he sets (12 h by default). Both actions check the slot again inside a transaction with a PostgreSQL advisory lock, so two people can never take the same slot.
- **Client booking link.** Created from the client file, it lets an existing client book a coaching session directly. The session counts against their package.
- **Notifications.** Sessions booked online get a badge in the calendar and the client file. Online bookings, moves and cancellations appear in the notification bell.

6 M4 · E-mail templates and sequences

- Settings › E-mails lists every automatic e-mail. Each one can be turned on or off and reworded (subject, title, text with variables such as `{{prenom}}`), with a live preview. Unknown variables are refused when saving.
- Events in the CRM schedule e-mails in a `scheduled_emails` table with a unique key, so one event never schedules the same e-mail twice:
  - a welcome e-mail when a contact becomes a client;
  - a follow-up the day after a discovery call, if the person has not become a client;
  - a message when one session is left in a package, plus a « propose a renewal » task for Jérémy;
  - a thank-you and testimonial request two days after a coaching ends.
- When an e-mail is due, the job checks that it still makes sense (for example, no follow-up if the person became a client), records why it was skipped, and retries a failed send up to three times.
- From a contact file, Jérémy can send an e-mail from a template or a blank message, and see or cancel the automatic e-mails that are planned.
- Contacts can stop the non-essential e-mails with a link at the bottom of those e-mails. Confirmations and reminders still go out.

7 M5 · Intake questionnaire

- The questions are editable in Settings: short or long answer, single or multiple choice, 1–10 scale, date.
- The link is sent in the welcome e-mail or from the client file. The page works well on a phone, accepts one submission per link and stays valid 30 days.
- Answers are saved with a copy of the questions, so changing the questionnaire later does not change old answers. They appear in a « Questionnaire » tab of the client file, and Jérémy gets an e-mail and a notification.

8 Testing

- Each step is a separate commit. Type check, lint and the production build pass.
- There are 66 automated tests. For the jobs and the links I test against a real PostgreSQL running in memory (PGlite). This lets me check things like « running the job twice sends nothing twice » or « a link can only be used once ».
- I also tested each feature by hand in the browser, in test mode (no real e-mails), and deleted the test data afterwards.
- Database changes are migrations 0005 to 0011, all additive.

9 Next steps

- **M6 · Invoices and payments:** waiting for Jérémy's decisions (country and VAT, Stripe account, payment before or after a session).
- **M7 · Dashboard:** revenue forecast, no-show trend, booking sources, and the final documentation.
- **Before going live:** a real sender address and DNS records on his domain, hosting on the SwissWAI server with the cron job, and a performance check of the slowest pages in a production build.